

Brainstorming & Idea Prioritization

Date	15 March 2024
Team Member Name	Naga Lakshmi
Project Name	Smart Lender (Modular Machine Learning Pipeline for Loan Eligibility Prediction)
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Naga Lakshmi	To build a machine learning pipeline using historical loan application data (Decision Tree, Random Forest, KNN, XGBoost) integrated into a Flask-based web application. The system will handle data cleaning, missing values, encoding, predict loan eligibility with a confidence score, support batch CSV prediction, and provide analytical dashboards for loan trends.	Fintech / Predictive Analytics	1

Step 2: Idea Prioritization

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	To build a modular machine learning pipeline for loan application processing with automated evaluations, Flask UI integration, batch prediction, and analytical monitoring dashboards.	High	High	High	Yes

Detailed System Architecture & Workflow

The Smart Lender system follows a modular machine learning pipeline. Historical loan application data is collected and preprocessed through data cleaning, missing value handling, categorical encoding, and feature transformation. The processed dataset is used to train multiple classification algorithms, including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost (or Logistic Regression, depending on the selected model). The trained models are evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The best-performing model is saved as a serialized model file and integrated into a Flask-based web application. During runtime, applicants submit their loan details through the user interface. The application preprocesses the input using the same pipeline, loads the trained model, predicts loan eligibility, calculates the confidence score, stores the prediction in the history database, and generates downloadable reports. The system also supports batch prediction using CSV files and provides analytical dashboards for monitoring loan approval trends and model performance.